reason? When all these objects are dropped in the absence of air medium (vacuum), all would have reached the ground at the same time. In air medium, air offers some resistance to the motion of freely falling objects. But, it is negligibly small when compared to the gravitational pull acting on the stone and rubber. Hence, they reach the ground at the same time.

It can be seen from these activities that the magnitude of air resistance depends on the area of objects exposed to air. We know that an object experiences acceleration during free fall. This acceleration experienced by an object is independent of mass. This means that all objects hollow or solid, big or small, should fall at the same rate.

The equation of motion for a freely falling body can be obtained by replacing 'a' in equations with g, the acceleration due to gravity. For a freely falling body which is initially at rest, u = 0. Thus we get the following equations.

$$v = gt,\ s = \tfrac{1}{2}\ gt^2,\ v^2 = 2gh$$

DO YOU KNOW?

Can a body have zero velocity and finite acceleration? Yes, when a body is thrown vertically upwards in space, at the highest point, the body has zero velocity but it has acceleration due to the gravity.

When we throw an object vertically upwards, it moves against the acceleration due to gravity. Hence, 'a' is taken to be –g and when moving downwards 'a' is taken as +g.

2.8 Uniform circular motion

Activity 5

Take a piece of thread and tie a small piece of stone at one of its ends. Rotate the stone to describe a circular path with constant speed by holding the thread at the other end. Now, release the thread and let the stone go. Can you tell the direction in which the stone moves after it is released?

If you carefully observe, on being released the stone moves along a straight line tangential to the circular path. This is because once the stone is released, it continues to move along the direction it has been moving at that instant. This shows that the direction of motion changes at every point when the stone was moving along the circular path.

When an object is moving with a constant speed along a circular path, the velocity changes due to the change in direction. Hence, it is an accelerated motion. For example, revolution of earth around the sun, revolution of moon around the earth and the tip of the second's hand of a clock are all accelarated motions.

If an object, moving along a circular path of radius, r takes time, T to come back to its starting position, then the speed, v is given by,

Speed = Circumference/Time taken

$$V = 2\pi r/T$$

2.9 Centripetal Acceleration and Centripetal Force

A body is said to be accelerated, if the velocity of the body changes either in magnitude or in direction. So, the motion of a stone in circular path with constant speed and continuous change of direction is an accelerated motion. In this case, there must be an acceleration acting along the string directed inwards, which makes the stone to move in circular path.

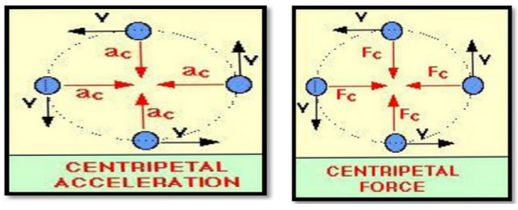


Figure 2.9 Centripetal acceleration and Centripetal force

This acceleration is known as centripetal acceleration and the force is known as centripetal force. Since the centripetal acceleration is directed radially towards the centre of the circle,

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